

COMS20018, Introduction to AI:

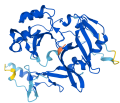
AI For X: A Landscape Tour

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The Landscape of AI Applications



Science



Finance



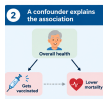
Generative AI



Recommenders



Game playing



Epidemiology



Computer vision



Conservation



Healthcare



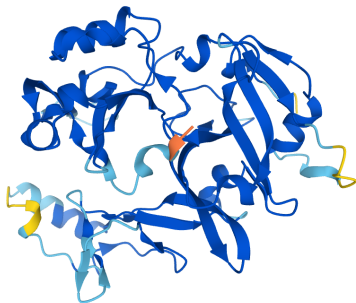
ChatGPT

Everyday AI tools

⇒ A quick tour of where AI already shows up around you, then how this unit gets you there.

Solving a 50-Year-Old Problem in Biology

AI for X · Science

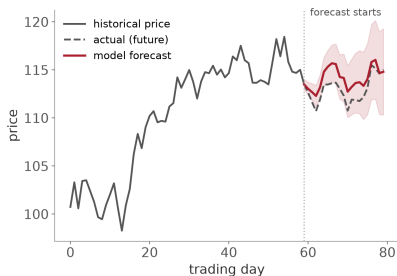


- ▶ The “protein folding problem”: predict a protein’s 3D shape from its amino-acid sequence
- ▶ AlphaFold 2 (DeepMind) predicts structures at near-experimental accuracy for most proteins
- ▶ **Under the hood:** a deep neural network trained on ~170,000 known structures
- ▶ 200+ million structures released freely; won the 2024 Nobel Prize in Chemistry

Image: human C12orf29 protein structure predicted by AlphaFold. CSBIOPASSION, Wikimedia Commons, CC BY-SA 4.0. Method: Jumper et al., *Nature* (2021).

Can a Model Predict Tomorrow's Price?

AI for X · Quantitative Finance



- ▶ Task: forecast a price (or a return) from historical patterns and market signals
- ▶ **Under the hood:** regression and recurrent/transformer models trained on historical time series
- ▶ The shaded band matters as much as the line; markets are noisy, so a good forecast reports uncertainty

Machines That Create: Images, Music, and More

AI for X · Generative Models



- ▶ “An astronaut riding a horse”; text prompt in, brand-new image out
- ▶ **Under the hood:** diffusion models start from pure noise and repeatedly refine it toward a coherent image, guided by the text prompt
- ▶ The same idea generates music (Suno, Udio) and video (Sora), not just images
- ▶ You’ll meet generative models in Week 3

Image: “Astronaut Riding a Horse”, generated with Stable Diffusion XL.

VulcanSphere, Wikimedia Commons, CC0 / public domain.

AI Tools You Already Use

AI for X · Everyday tools



ChatGPT



Claude



Gemini



Copilot

- ▶ These are large language models (LLMs): transformer neural networks with billions of parameters
- ▶ Trained in two stages:
 - ▶ self-supervised pretraining on huge amounts of text, predict the next word
 - ▶ fine-tuning with human feedback (RLHF) to be helpful, honest, and safe
- ▶ You'll learn the Transformer and LLM behind these in Week 3 and 4

Can AI Ever Be Conscious?

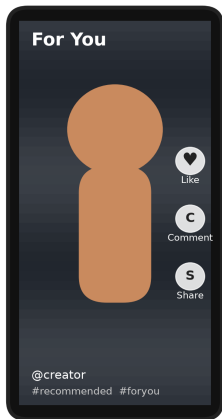
AI for X · Philosophy of AI

- ▶ LLMs write fluently, hold conversations, and sometimes claim to have feelings; does that mean anything is actually happening “inside”?
- ▶ A live scientific and philosophical debate, not one that better benchmarks alone will settle

Birhane, A. “Can AI ever be conscious? The question stems from a misconception.” *Nature*, 656 (2026). <https://www.nature.com/articles/d41586-026-02571-9>

How TikTok Decides What's Next

AI for X · Recommender Systems

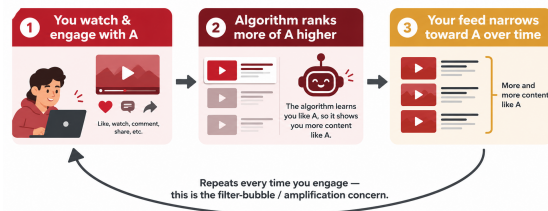


- ▶ Task: predict which video you'll watch (and for how long), from your past behaviour
- ▶ **Signals used:** watch time, likes, comments, shares, feed a ranking model that scores every candidate video for you
- ▶ **Under the hood:** deep learning embeddings of users and videos, updated continuously as you interact, a modern descendant of matrix factorisation

See TikTok's public "How TikTok recommends videos" (2020).

Biased Recommendation and Exploration

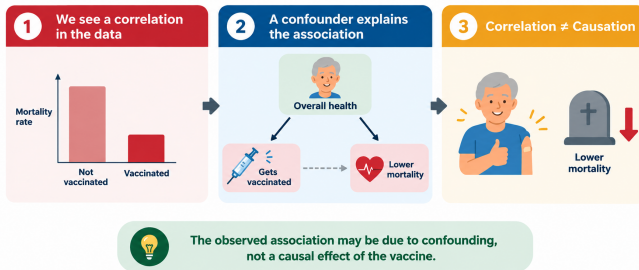
AI for X · Ethics & Reinforcement Learning



- ▶ Recommenders are optimised for engagement (watch time, clicks), not for your wellbeing or a balanced information diet
- ▶ The same feedback loop that makes recommendations good also makes bias self-reinforcing: filter bubbles, algorithmic amplification
- ▶ **The explore/exploit trade-off:** always showing what you already like (*exploit*) entrenches the bias; trying something different (*explore*) is how the system learns (Week 8 multi-armed bandits)

Correlation Isn't Causation

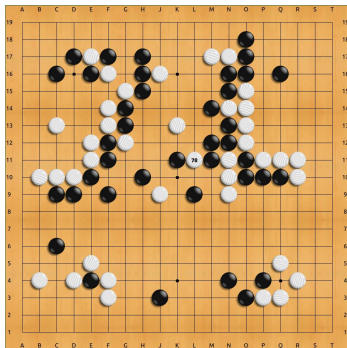
AI for X · Epidemiology



- ▶ Do healthier people simply choose to get vaccinated more often?
- ▶ Machine learning finds patterns; causal inference asks whether a pattern reflects a real effect
- ▶ This distinction, correlation vs. causation, is the subject of Week 8

Beating the World Champion at Go

AI for X · Reinforcement Learning

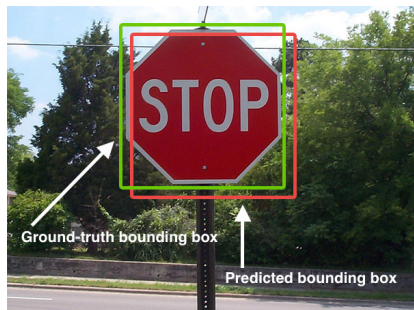


- ▶ **The game of Go has more possible positions than atoms in the universe**, too many to search exhaustively
- ▶ AlphaGo (DeepMind, 2016) beat world champion Lee Sedol 4–1
- ▶ **Under the hood:** trained by playing millions of games against itself, learning from the reward of winning or losing (week 9: reinforcement learning)
- ▶ Same family of ideas behind game-playing agents, robotics, and RLHF fine-tuning of LLMs

Image: board position after AlphaGo's "divine move" 78 vs. Lee Sedol (2016). Axd, Wikimedia Commons, CC BY-SA 4.0.

Teaching Machines to See

AI for X · Computer Vision

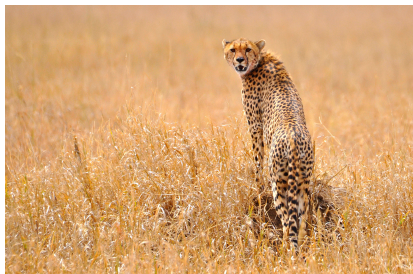


- ▶ Task: locate and label objects in an image (detection), or classify a whole image (classification)
- ▶ **Under the hood:**
Convolutional Neural Networks (CNNs) trained via supervised learning on millions of labelled images
- ▶ Applications: self-driving cars, quality control, satellite imagery

Image: Adrian Rosebrock, PyImageSearch, Wikimedia Commons, CC BY-SA 4.0.

Spotting Endangered Species From Millions of Photos

AI for X · Computer Vision, continued

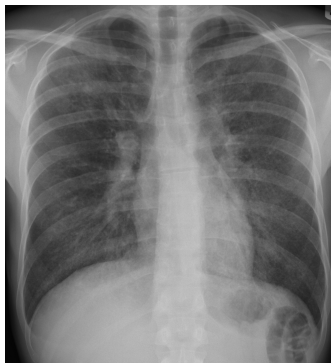


- ▶ Camera traps and drones collect far more wildlife photos than ecologists can label by hand
- ▶ **Under the hood:** CNNs classify species (and count individuals) automatically, at a scale no human team could match
- ▶ Used by conservation projects (e.g. Wildlife Insights, Snapshot Serengeti) to track endangered populations

Image: cheetah, Serengeti National Park. Tobi 87, Wikimedia Commons, CC BY-SA 3.0.

A Second Opinion From a Neural Network

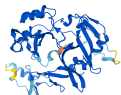
AI for X · Healthcare



- ▶ CNNs trained on labelled X-rays can flag disease (e.g. pneumonia) at radiologist-level accuracy
- ▶ **Example:** CheXNet, a 121-layer CNN trained on 100,000+ chest X-rays (Stanford, 2017)
- ▶ These are decision-support tools, not replacements for clinicians
- ▶ **Ties to Trustworthy AI (Week 11):** safety, fairness and accountability matter enormously here

Image: chest X-ray showing *Pneumocystis* pneumonia. Samir, Wikimedia Commons, public domain. Method: Rajpurkar et al., CheXNet (2017).

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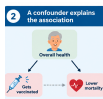
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